Snare Locations

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Contents

Formatting Snare Locations

Here, we will create spatial objects with information on the location of each of the J hair snares (*i.e.*, traps) associated with the project. There are several input files needed...

1. `input/Hair Snare Summaries (Both Years).xlsx` - an Excel spreadsheet with information on landowners, locations, and weekly check dates for all hair snares in each of the two years of the project
2. `input/HS23_coords.csv` - a handful of columns from the 2023 tab of the file above (Site Name, Landowner, Latitude, Longitude, Date Installed, Date Baited)
3. `input/HS24_coords.csv` - a handful of columns from the 2024 tab of the file above (Site Name, Landowner, Latitude, Longitude, Date Installed, Date Baited)

The primary outputs will include...

1. `1_snareLocations.pdf` - a markdown (in PDF format) that walks through the steps
2. `snareLocations.Rda` - an `.Rda` file with two objects
    - `locs` - coordinates for all unique hair snares associated with the project (a $J \times 3$ data frame)
    - `oper` - operational status of each hair snare in each of the two years (a $J \times 2$ data frame)
3. `translatable.csv` - a table with translation information for the naming of snares

Load files

Read the `.csv` files with coordinates into R. Remove rows associated with snares *NOT* operated in a given year (these will have no information - " " - in the `Date.Baited` column)...

```r
HS23 <- read.csv("../input/HS23_coords.csv", header = TRUE)
HS24 <- read.csv("../input/HS24_coords.csv", header = TRUE)
sum(!is.na(HS23$Longitude) & HS23$Date.Baited != "")
```

```
## [1] 130
```

```r
sum(!is.na(HS24$Longitude) & HS24$Date.Baited != "")
```

```
## [1] 138
```

```r
HS23 <- HS23[HS23$Date.Baited != "",]
HS24 <- HS24[HS24$Date.Baited != "",]

dim(HS23)
```

```
## [1] 130   6
```

```r
dim(HS24)
```

```
## [1] 138   6
```

Convert to UTM

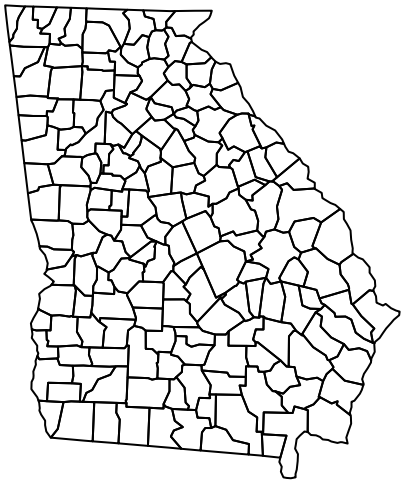
Now convert to UTM projection, which we'll use for all spatial objects.

Note that `crs=32617` is the coordinate reference system for UTM zone 17N (between 78°W and 84°W, our study region sits around 83.5°W) and `crs=4326` is for WGS84 lat/longs.

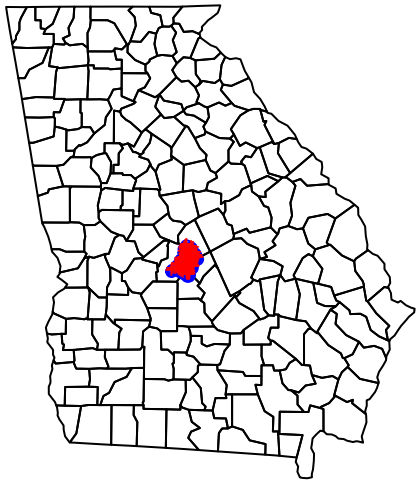
```r
library(sf)
```

```
## Linking to GEOS 3.13.1, GDAL 3.11.0, PROJ 9.6.2; sf_use_s2() is TRUE
```

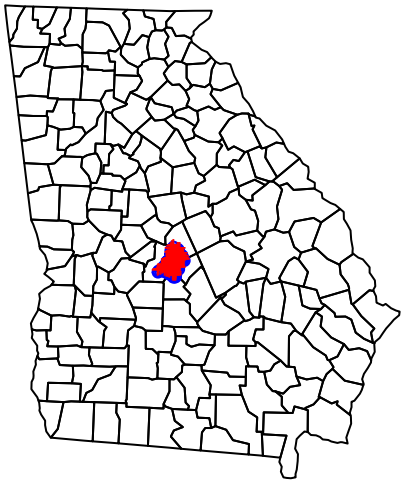
```r
library(usmap)

#Create a UTM map of GA counties using the usmap R package
gamap <- us_map(regions="counties", include="GA")
gamap <- st_transform(gamap, crs=4326) #Convert to WGS84
gamap_utm <- st_transform(gamap, crs=32617) #Convert to UTM
plot(st_geometry(gamap_utm))
```



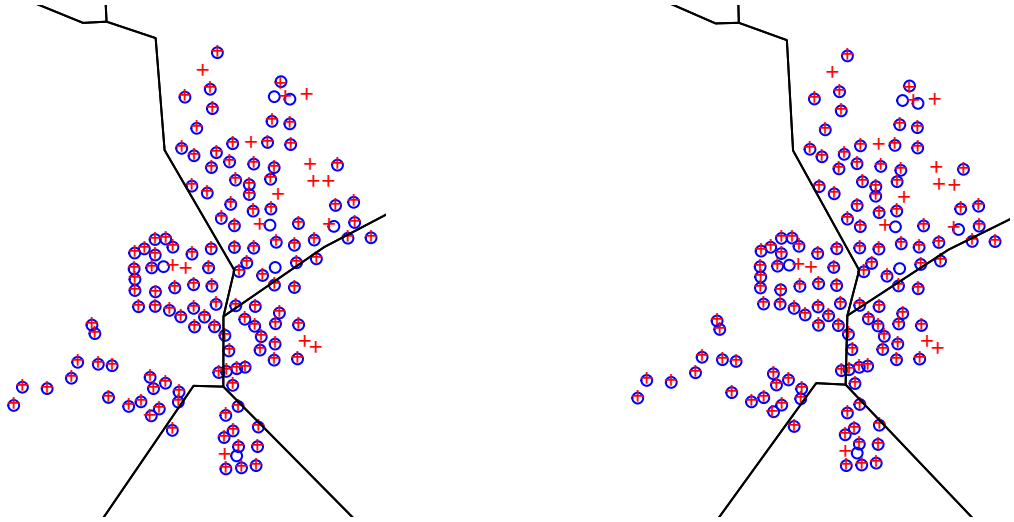
```r
#Convert to sf objects
hs23_sf <- st_as_sf(x=HS23[,c(4,3)], coords = c("Longitude","Latitude"), crs= 4326)
hs24_sf <- st_as_sf(x=HS24[,c(4,3)], coords = c("Longitude","Latitude"), crs= 4326)
plot(st_geometry(gamap))
plot(st_geometry(hs23_sf), pch=1, col="blue", add=TRUE, cex = 0.75)
plot(st_geometry(hs24_sf), pch="+", col="red", add=TRUE, cex = 0.75)
```



```
#Transform to UTM coordinates
#32617 is the epsg for UTM zone 17N
hs23_utm <- st_transform(hs23_sf, crs=32617)
hs24_utm <- st_transform(hs24_sf, crs=32617)

#Check the plot
plot(st_geometry(gamap_utm))
plot(st_geometry(hs23_utm), pch=1, col="blue", add=TRUE, cex = 0.75)
plot(st_geometry(hs24_utm), pch="+", col="red", add=TRUE, cex = 0.75)
```



```
#Plot these objects in reverse order to double check fine-scale positioning
#And with WGS and UTM next to one another to double check everything
par(mfrow = c(1,2))
#WGS84
plot(st_geometry(hs23_sf), pch=1, col="blue", cex = 0.75)
plot(st_geometry(hs24_sf), pch="+", col="red", add=TRUE, cex = 0.75)
plot(st_geometry(gamap), add=TRUE)
#UTM
plot(st_geometry(hs23_utm), pch=1, col="blue", cex = 0.75)
plot(st_geometry(hs24_utm), pch="+", col="red", add=TRUE, cex = 0.75)
plot(st_geometry(gamap_utm), add=TRUE)
```



Now let's create some unique names (based on UTM coordinates, as in Hooker et al. (2020)) for the traps.

```
#Create unique names based on the UTMcoords for each trap
tmp23 <- st_coordinates(hs23_utm)
names23 <- paste0(round(tmp23[,"X"]),"x",round(tmp23[,"Y"]))
rownames(tmp23) <- names23
tmp24 <- st_coordinates(hs24_utm)
names24 <- paste0(round(tmp24[,"X"]),"x",round(tmp24[,"Y"]))
rownames(tmp24) <- names24
allnames <- unique(c(names23,names24))
length(allnames)
```

```
## [1] 154
```

```
traplocs <- data.frame(East=rep(NA,length(allnames)), North=rep(NA,length(allnames)))
oper <- data.frame(Yr2023=rep(NA,length(allnames)), Yr2024=rep(NA,length(allnames)))
rownames(traplocs) <- allnames
rownames(oper) <- allnames

#Check to make sure there are never mismatches in locations when the UTM name matches
bothyrs <- which(allnames %in% rownames(tmp23) & allnames %in% rownames(tmp24))
checknames <- allnames[bothyrs]
cnt <- 0 #Counter for matches
for(x in 1:length(checknames)) {
    if(tmp23[,"X"][rownames(tmp23) == checknames[x]] ==
       tmp24[,"X"][rownames(tmp24) == checknames[x]]
```

```
       & tmp23[,"Y"][rownames(tmp23) == checknames[x]] ==
       tmp24[,"Y"][rownames(tmp24) == checknames[x]]) {
        cnt <- cnt+1 #Add 1 to counter if the X and Y coords & trap name match
    }
}
length(checknames)
```

```
## [1] 114
```

```
cnt
```

```
## [1] 114
```

```
#So they all match up perfectly... good
```

Create `oper` and `traplocs`

Now fill in the `oper` and `traplocs` objects that will be used for modeling.

```
#Now fill in the contents of traplocs and oper
for(j in 1:length(allnames)) {
  #If the trap name is in both years...
    if(allnames[j] %in% rownames(tmp23) & allnames[j] %in% rownames(tmp24)) {
        traplocs[j,] <- tmp23[which(rownames(tmp23) == allnames[j]),]
        oper[j,] <- c(1,1)
    #If only on 2023
    } else if (allnames[j] %in% rownames(tmp23)) {
        traplocs[j,] <- tmp23[which(rownames(tmp23) == allnames[j]),]
        oper[j,] <- c(1,0)
    #If only in 2024
    } else if(allnames[j] %in% rownames(tmp24)) {
        traplocs[j,] <- tmp24[which(rownames(tmp24) == allnames[j]),]
        oper[j,] <- c(0,1)
    }
}

#Now check your work...
head(traplocs)
```

```
##                    East   North
## 248187x3586614 248187.0 3586614
## 255814x3587061 255814.0 3587061
## 248920x3588050 248919.6 3588050
## 250904x3587890 250904.0 3587890
## 252875x3588670 252874.7 3588670
## 253409x3589929 253409.2 3589929
```

```
head(oper)
```

```
##                 Yr2023 Yr2024
## 248187x3586614      1      1
## 255814x3587061      1      1
## 248920x3588050      1      1
## 250904x3587890      1      1
## 252875x3588670      1      1
## 253409x3589929      1      1
```

```
#These should be 0... no rownames shared when oper == 0
sum(rownames(traplocs)[oper$Yr2023 == 0] %in% rownames(tmp23))
```

```
## [1] 0
```

```
sum(rownames(traplocs)[oper$Yr2024 == 0] %in% rownames(tmp24))
```

```
## [1] 0
```

```
#Checking the length of the objects vs. sum of oper and matching rownames
#2023
length(HS23$Site.Name)
```

```
## [1] 130
```

```
sum(oper$Yr2023)
```

```
## [1] 130
```

```
sum(rownames(traplocs)[oper$Yr2023 == 1] %in% rownames(tmp23))
```

```
## [1] 130
```

```
#2024
length(HS24$Site.Name)
```

```
## [1] 138
```

```
sum(oper$Yr2024)
```

```
## [1] 138
```

```
sum(rownames(traplocs)[oper$Yr2024 == 1] %in% rownames(tmp24))
```

```
## [1] 138
```

Create `translatable`

Now create an object that we can use to translate between the various names used for hair snares in the dataset.

```r
#Now create translatable
translatable <- data.frame(UTMname=rep(NA, length(allnames)),
                           name23 = rep(NA,length(allnames)),
                           name24 = rep(NA, length(allnames)),
                           oper23 = rep(NA,length(allnames)),
                           oper24 = rep(NA,length(allnames)),
                           lat=rep(NA,length(allnames)),
                           long=rep(NA,length(allnames)),
                           east=rep(NA,length(allnames)),
                           north=rep(NA,length(allnames)))
translatable$UTMname <- allnames
for(x in 1:length(allnames)) {
  tmpcoords23 <- tmpcoords24 <- c()
    if(allnames[x] %in% rownames(tmp23)) {
        translatable$name23[x] <- HS23$Site.Name[which(rownames(tmp23) == allnames[x])]
        tmpcoords23 <- c(HS23$Latitude[which(rownames(tmp23) == allnames[x])],
                         HS23$Longitude[which(rownames(tmp23) == allnames[x])])
    }
    if(allnames[x] %in% rownames(tmp24)) {
        translatable$name24[x] <- HS24$Site.Name[which(rownames(tmp24) == allnames[x])]
        tmpcoords24 <- c(HS24$Latitude[which(rownames(tmp24) == allnames[x])],
                         HS24$Longitude[which(rownames(tmp24) == allnames[x])])
    }
  translatable$east[x] <- traplocs[which(rownames(traplocs) ==
                                           translatable$UTMname[x]),"East"]
  translatable$north[x] <- traplocs[which(rownames(traplocs) ==
                                            translatable$UTMname[x]),"North"]
  if(length(tmpcoords23) > 0 & length(tmpcoords24) > 0) {
    if(sum(tmpcoords23 == tmpcoords24)==2) {
      translatable$lat[x] <- tmpcoords23[1]
      translatable$long[x] <- tmpcoords23[2]
    } else {
      stop(paste("Error for",allnames[x],"coords in 2023 and 2024 don't match!!"))
    }
  } else {
    if(length(tmpcoords23) > 0) {
      translatable$lat[x] <- tmpcoords23[1]
      translatable$long[x] <- tmpcoords23[2]
    } else if(length(tmpcoords24) > 0) {
      translatable$lat[x] <- tmpcoords24[1]
      translatable$long[x] <- tmpcoords24[2]
    }
  }
  translatable$oper23[x] <- oper[which(rownames(oper)==allnames[x]),1]
  translatable$oper24[x] <- oper[which(rownames(oper)==allnames[x]),2]
  rm(tmpcoords23, tmpcoords24)
}

head(translatable)
```

```
##          UTMname name23 name24 oper23 oper24      lat      long     east
## 1 248187x3586614 HS 001 HS 001      1      1 32.38816 -83.67687 248187.0
## 2 255814x3587061 HS 002 HS 002      1      1 32.39388 -83.59598 255814.0
```

```
## 3 248920x3588050 HS 003 HS 003      1      1 32.40127 -83.66947 248919.6
## 4 250904x3587890 HS 004 HS 004      1      1 32.40027 -83.64835 250904.0
## 5 252875x3588670 HS 005 HS 005      1      1 32.40774 -83.62762 252874.7
## 6 253409x3589929 HS 006 HS 006      1      1 32.41920 -83.62227 253409.2
##     north
## 1 3586614
## 2 3587061
## 3 3588050
## 4 3587890
## 5 3588670
## 6 3589929
```

```
#A few checks before saving the output so far...
#Any cases where the name doesn't match (i.e., same name, different trap?)
sum(!translatable$name23 == translatable$name24, na.rm = TRUE)
```

```
## [1] 0
```

```
#How many traps not operated in both years
sum(is.na(translatable$name23 == translatable$name24))
```

```
## [1] 40
```

```
table(rowSums(oper)) #114 traps remain in the same spot, 40 are new or moved
```

```
##
##   1   2
##  40 114
```

```
#There are three options, 10, 01, and 11 for operation over the 2 years
table(apply(oper,1,function(x) paste0(x,collapse="")))
```

```
##
##  01  10  11
##  24  16 114
```

```
#114 traps operated in both years, 16 in 2023 but not 2024, 24 in 2024 but not 2023

locs <- traplocs
translatable$name24 <- gsub("B","",translatable$name24)

write.csv(translatable, "translatable.csv", quote=FALSE, row.names=FALSE)
save(oper, locs, file="snareLocations.Rda")
```

Visualize the snare locations

Plot the traps and superimpose a 4km buffer (the size of the starting buffer in Hooker et al. (2020)). This buffer will not be used as the state space for subsequent analyses, but it does serve to illustrate the potential locations of activity centers in relation to the hair snares used in the study.

```
library(shapefiles)
```

```
## Loading required package: foreign
```

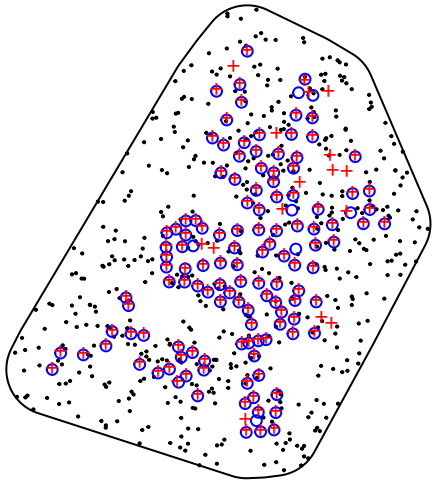
```
##
## Attaching package: 'shapefiles'
```

```
## The following objects are masked from 'package:foreign':
##
##     read.dbf, write.dbf
```

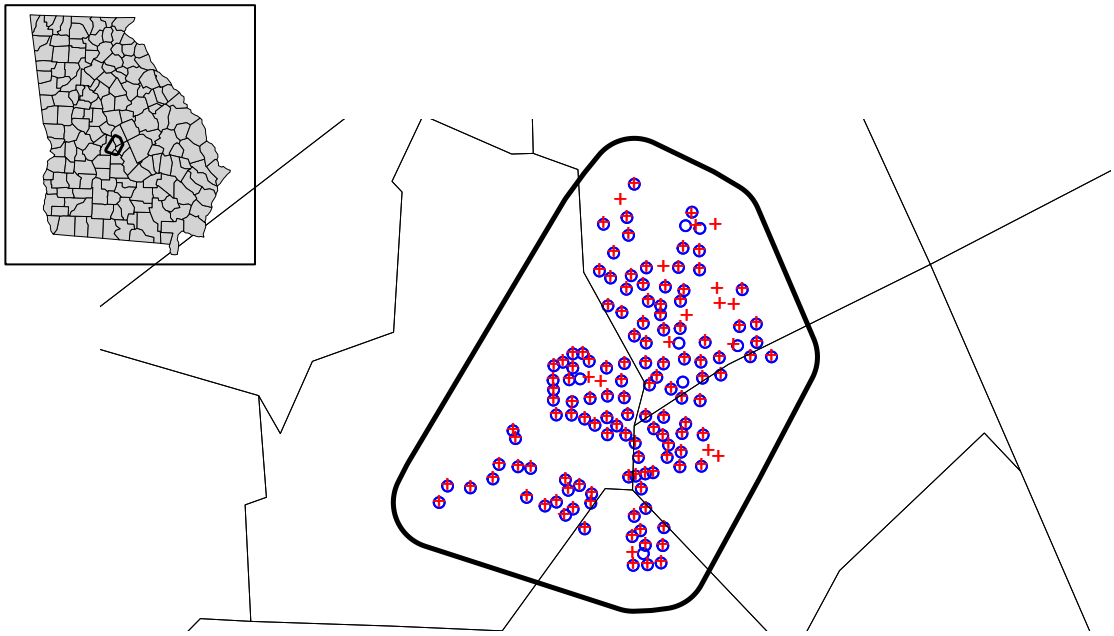
```
buffer <- 4000 #Buffer setting, in meters
tr <- traplocs

#Make a polygon from the trap locations
p1 <- st_polygon(list(as.matrix(rbind(tr,tr[1,]))))
#identify the points that form the convex hull of p1
p1ch <- p1[[1]][chull(p1[[1]]),]
p1ch_closed <- st_polygon(list(as.matrix(rbind(p1ch,p1ch[1,]))))

#Add the buffer to the convex hull
bp1 <- st_buffer(p1ch_closed,buffer)

#Extract coordinates from the st object and build a shapefile
bp1.coords <- bp1[[1]]
shpdf <- data.frame(Id=rep(1,nrow(bp1.coords)), X=bp1.coords[,1], Y=bp1.coords[,2])
shptbl <- data.frame(Id = 1)
shpfl <- convert.to.shapefile(shpdf, shptbl, "Id", 5)
system("mkdir CGAsnarebuff")
write.shapefile(shpfl, "CGAsnarebuff/CGAsnarebuff")
CGA <- st_read("CGAsnarebuff")
```

```
## Reading layer 'CGAsnarebuff' from data source
##   '/scratch/jdrobins/CGA_SCR/sampleinfo/CGAsnarebuff' using driver 'ESRI Shapefile'
## Simple feature collection with 1 feature and 1 field
## Geometry type: POLYGON
## Dimension:     XY
## Bounding box:  xmin: 244187 ymin: 3577135 xmax: 281169.6 ymax: 3618369
## CRS:           NA
```

```
plot(st_geometry(CGA))
S1 <- st_sample(CGA, size = 500)
plot(S1, add=TRUE, pch = 19, cex=0.15)
points(traplocs[oper[,1] == 1,], pch=1, col="blue", cex=0.75)
points(traplocs[oper[,2] == 1,], pch="+", col="red", cex=0.75)
```



```r
#Just as a reminder, here's the original trap array with county lines included
#Shouldn't be able to tell that points are overlaid here...
plot(st_geometry(CGA), lwd = 2.5)
plot(st_geometry(hs23_utm), pch=1, col="blue", add=TRUE, cex = 0.75)
points(traplocs[oper[,1] == 1,], pch=1, col="blue", cex=0.75)
plot(st_geometry(hs24_utm), pch="+", col="red", add=TRUE, cex = 0.75)
points(traplocs[oper[,2] == 1,], pch="+", col="red", cex=0.75)
plot(st_geometry(gamap_utm), lwd = 0.25, add=TRUE)
par(fig=c(0.05,0.25,0.65,0.95), mar=c(0,0,0,0), new=TRUE)
plot(st_geometry(gamap_utm), col="light grey", lwd = 0.25)
plot(st_geometry(CGA), add=TRUE, lwd = 1.5)
box(which="figure", lwd=1)
```



So at this point we have our `traplocs` and `oper` objects (which will be used directly in the SCR model). Now we can move on to matching the samples back to the hair snares where they were collected.
